Four Linux operating modes

Administrators select one or more modes and their order to fit the environment.

Invoke craft as the user

Real UID, NSS name and home
No arguments; no sudo

Read administrator-controlled mode order

Example: modes=home key_trust enrollment constrained_delegation

Try the next enabled mode; stop at the first success

A reported acquisition failure advances through the list. Mode numbers identify workflows, not priority.

1 Enrollment

enrollment
Service keytab + agent + CES
CA issues a user certificate
PKINIT obtains a user TGT
CA restricts recipients

2 Key Trust

key_trust
Service keytab + writable DC
Link a temporary public key
PKINIT obtains a user TGT
Remove the linked key

3 Windows PEM

home
Caller validates home PEM pair
PKINIT obtains a user TGT
KDC maps certificate to user
Supplied PEM pair persists

4 Service tickets

constrained_delegation
Dedicated broker AES keytab
S4U2Self then S4U2Proxy
AD restricts target services
No user or broker TGT export

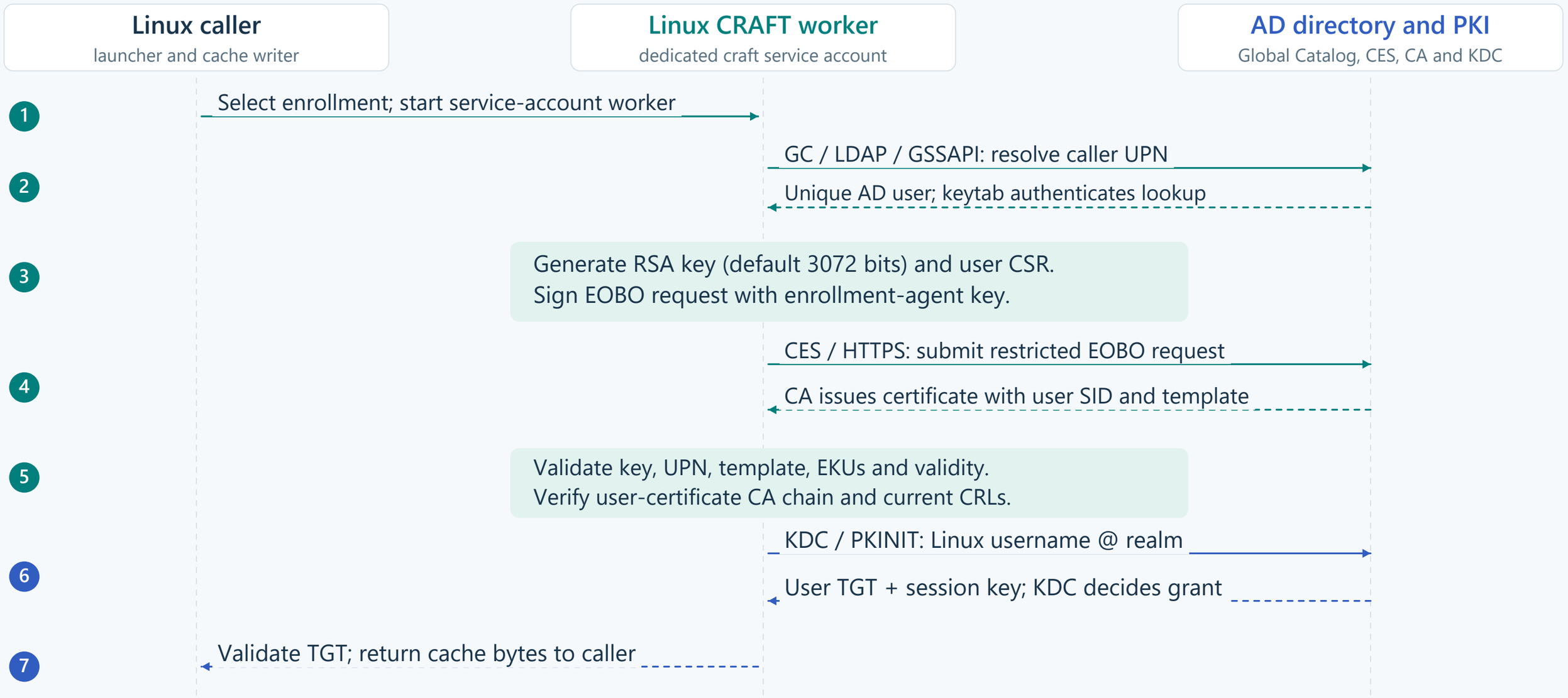
Validate credentials and atomically publish as the caller

Linux username @ realm; .krb5cc_craft mode 0600; select with KRB5CCNAME

SOURCE source/README.md: Mode selection and mode-specific setup; docs/05-Constrained-Delegation.md

Mode 1 Privileged certificate enrollment

Linux orchestrates CA enrollment first, then PKINIT. Select modes=enrollment to require this workflow.

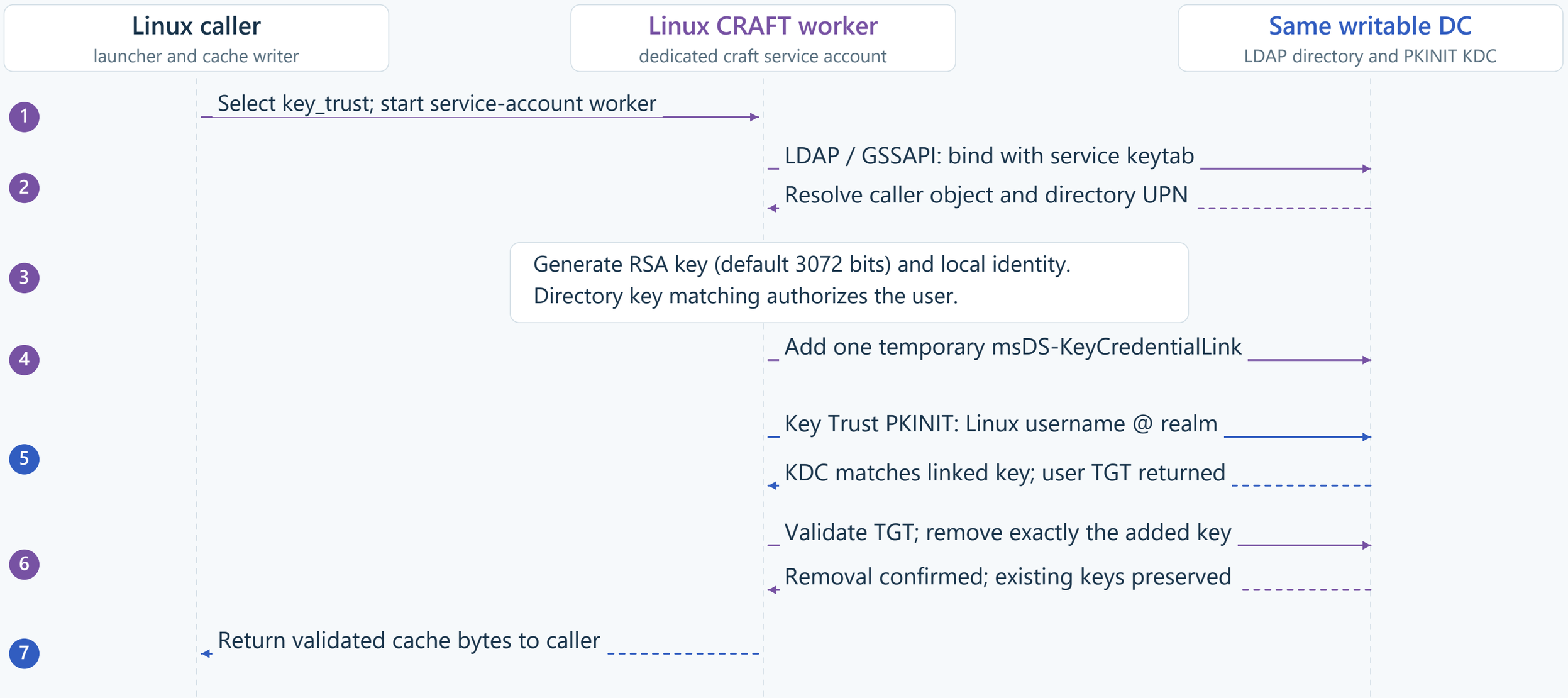


Caller publishes .krb5cc_craft mode 0600. Temporary user key and certificate are released.

SOURCE source/README.md: Mode 1 setup; docs/03-Enrollment-Agent-Restrictions.md

Mode 2 Privileged credential linking

Linux links a temporary AD public key, performs Key Trust PKINIT, and removes the entry before publishing the TGT.



Caller publishes .krb5cc_craft mode 0600. Cleanup also runs after acquisition failure.
Failed cleanup blocks publication and reports a residual key for administrator removal.

SOURCE source/README.md: Mode 2 setup; docs/04-Key-Credential-Link-Delegation.md

Mode 3 Unprivileged Windows-exported certificate

Windows enrolls and exports the certificate. Linux uses the transferred PEM pair entirely as the caller.

Windows domain user
direct user enrollment and export

Linux user and CRAFT
ordinary executables; caller UID

AD CS and KDC
certificate issuance and PKINIT

1. Windows enrollment policy: request a user certificate with an exportable software key →
   ← AD CS issues the user certificate; private key stays with the user
2. Request-CRAFT-Certificate.cmd exports user.pem + matching unencrypted user.key
3. Securely transfer both PEM files to Linux →
4. Install at NSS home: ~/.config/craft/user.pem + user.key
   Run craft as the user; check files and certificate.
   Validate CA chain and CRLs; use public /etc/craft policy.
5. KDC / PKINIT: Linux username @ realm →
   ← KDC maps certificate and returns user TGT
6. Validate returned TGT; publish mode-0600 .krb5cc_craft.
   PEM files remain; replace the pair before expiry.

No Linux setuid bit, service account, directory keytab, LDAP lookup, CES or /run/craft.

SOURCE source/README.md: Mode 3 setup; scripts/Request-CRAFT-Certificate.cmd

Ticket clocks and certificate storage

Modes 1-3 request 10 h / 7 d by default. Certificate lifetime and KDC policy constrain the initial grant.

1 / ENROLLMENT
Temporary key and certificate.
CA issuance records remain.

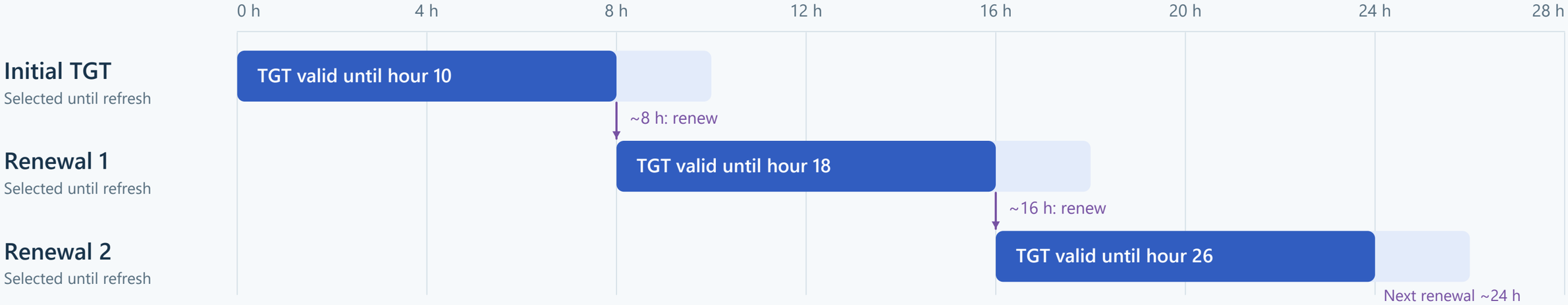
2 / CREDENTIAL LINKING
Temporary key and local identity.
Remove directory entry after PKINIT.

3 / WINDOWS-EXPORTED PEM
Persistent supplied certificate and key.
Replace the pair before expiry.

01 / HOURS

Renew before expiry

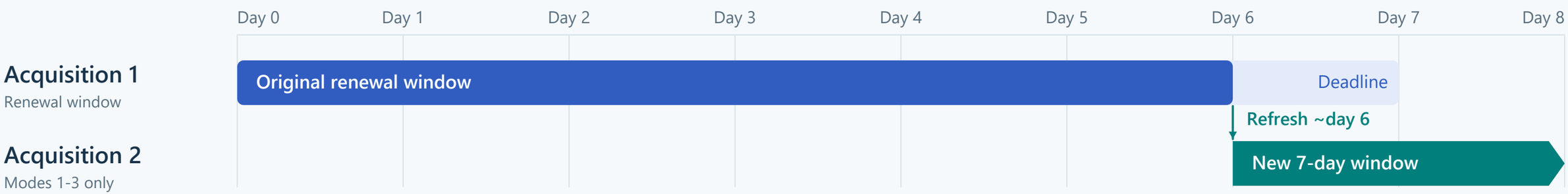
Illustrative full 10 h grants



Renewal advances expiry; renew-till stays fixed.

02 / DAYS

Start a fresh cycle near the renewal deadline



Renewal uses the cached ticket and session key. Certificate expiry or cleanup does not revoke an issued ticket.

SOURCE source/README.md: Configuration reference; Long-running jobs

Keep a job authenticated, then stop with it.

craft-maintain runs as the caller, coordinates one shared cache per UID, and watches a live process.

START ONCE / WATCH A SHELL THAT STAYS ALIVE FOR THE JOB

```
job_pid=$$
cache=$(/usr/local/bin/craft-maintain --watch-pid "$job_pid") || exit "$?"
export KRB5CCNAME="$cache"
```

Returns when usable credentials and background maintenance are ready.

01 / RENEW

Use the current ticket

When a renewable TGT has <=20% of its lifetime left (capped at 2 h), and its expiry can extend.

A renewal error retries; no immediate fresh authentication.

Cached TGT + session key → **KDC** renew request → **Validated TGT** new expiry

Renew-till never moves later.
Retain usable service tickets.

02 / FRESH AUTHENTICATION

Obtain fresh credentials

Missing/expired credentials or TGT rollover; mode 4: the earliest service ticket is near expiry.

Rollover margin: 1/7 of the renewal window, capped at 1 day.

Run craft select source → **Selected mode** 1, 2, 3 or 4 → **KDC** TGT or services

PKINIT or broker constrained delegation.
Replace the complete cache after validation.

A due renewal takes priority during rollover, keeping usable credentials alive while fresh authentication is delayed.

Successful refresh

Publish atomically to .krb5cc_craft (caller-owned, mode 0600).

Refresh failure

Keep the existing cache and retry; the job continues.

RETRY

Renewal: ~30 s to 5 min + jitter.
Fresh acquisition: 1 min to 1 h cooldown.

STOP

Watched PID exits, duration ends or signal.
Maintainer stops; the shared cache remains.

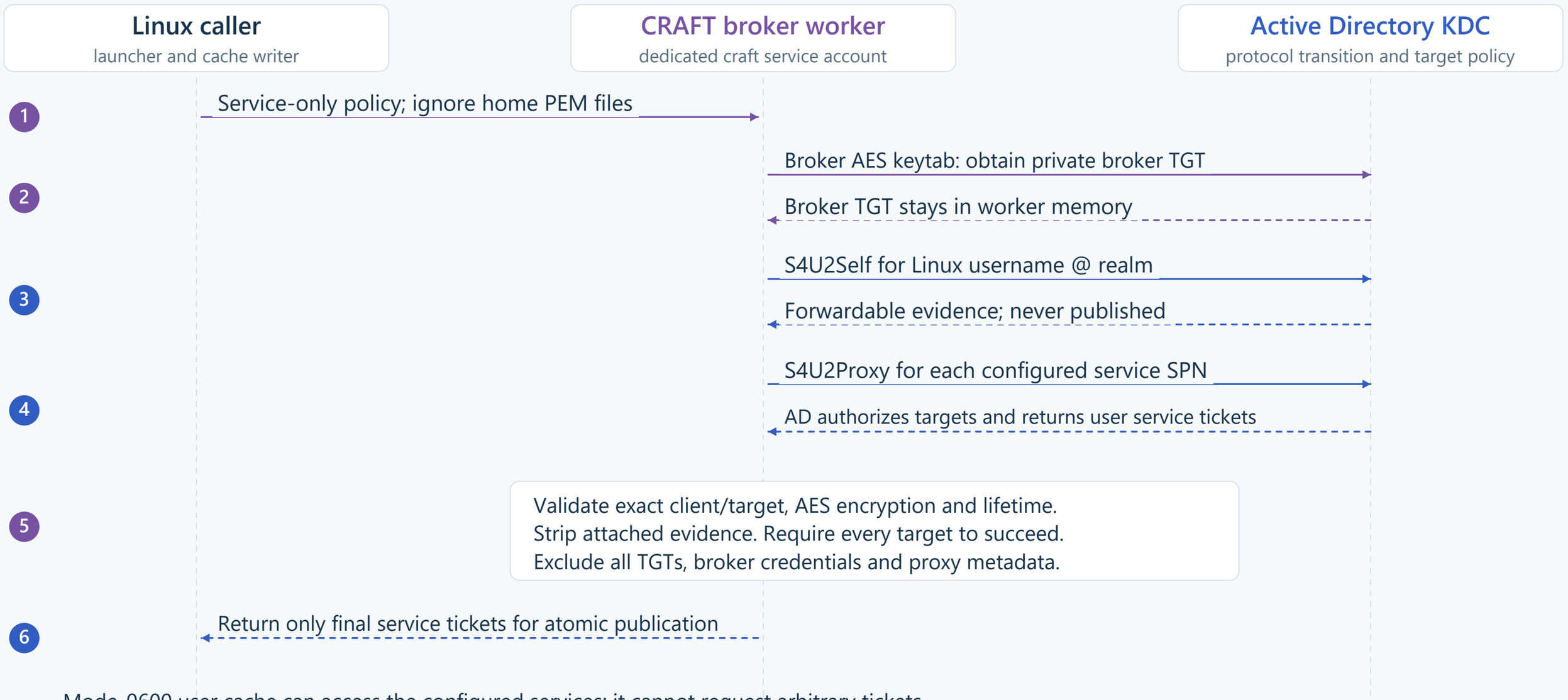
APPLICATION

Reload credentials when authenticating again.
Access follows the user's service permissions.

SOURCE source/README.md: Long-running jobs; source/SECURITY.md: Immediate disable

Mode 4 Service tickets through constrained delegation

The broker requests approved services for the caller. No user TGT or broker credentials leave the worker.



Mode-0600 user cache can access the configured services; it cannot request arbitrary tickets.
Maintenance refreshes the whole batch through the broker before the earliest service expiry.

SOURCE source/README.md: Mode 4 setup; docs/05-Constrained-Delegation.md